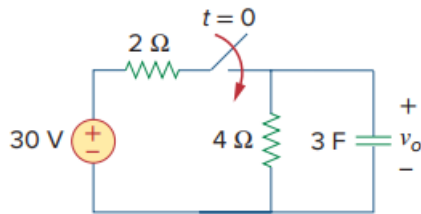


Problem

- (a) If the switch in Fig. 7.109 has been open for a long time and is closed at $t = 0$, find $v_o(t)$.
 (b) Suppose that the switch has been closed for a long time and is opened at $t = 0$. Find $v_o(t)$.

Fig. 7.109:



Step-by-step solution

Step 1 of 11

Refer to Figure 7.109 given in the textbook.

(a)

Consider that the switch is opened for a long time. Redraw the circuit for $t < 0$ as shown in Figure 1.

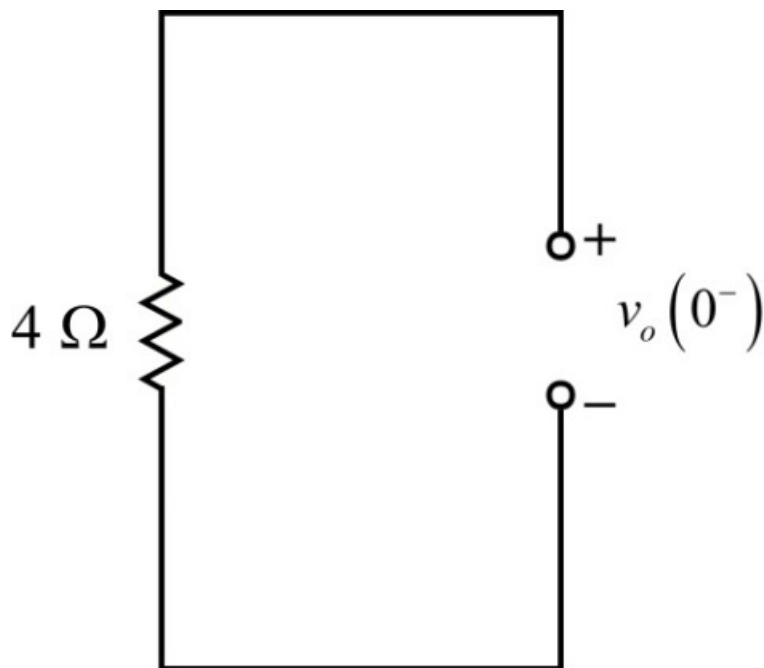


Figure 1

Step 2 of 11

Calculate the initial voltage $v_o(0)$ across the capacitor.

$$v_o(0^-) = v_o(0^+) = v_o(0) = 0 \text{ V}$$

Step 3 of 11

Consider that the switch is closed at $t = 0$. Redraw the circuit for $t > 0$ as shown in Figure 2.

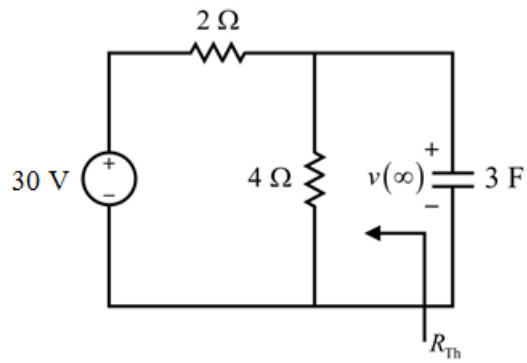


Figure 2

Step 4 of 11

Calculate the steady state voltage $v(\infty)$ across the capacitor.

Apply voltage division principle.

$$\begin{aligned} v(\infty) &= \frac{4}{4+2}(30) \\ &= \frac{4}{6}(30) \\ &= 20 \text{ V} \end{aligned}$$

Step 5 of 11

The time constant τ is,

$$\tau = R_{Th} C \dots\dots (1)$$

Calculate the Thevenin's resistance R_{Th} shown in Figure 2, with voltage source assumed as short circuit.

$$\begin{aligned} R_{Th} &= \frac{2 \times 4}{2+4} \\ &= \frac{8}{6} \\ &= \frac{4}{3} \Omega \end{aligned}$$

Determine the time constant τ .

Substitute $\frac{4}{3} \Omega$ for R_{Th} , and 3F for C.

$$\begin{aligned} \tau &= \frac{4}{3} \times 3 \\ &= 4 \text{ s} \end{aligned}$$

Step 6 of 11

The expression for the voltage across the capacitor is,

$$v_o(t) = v(\infty) + [v(0) - v(\infty)]e^{\frac{-t}{\tau}}$$

Determine the voltage $v_o(t)$ across the capacitor.

Substitute 0V for $v(0)$, 20V for $v(\infty)$, and 4s for τ .

$$v_o(t) = v(\infty) + [v(0) - v(\infty)]e^{\frac{-t}{\tau}}$$

$$v_o(t) = 20 + [0 - 20]e^{\frac{-t}{4}}$$

$$= 20 - 20e^{\frac{-t}{4}} \text{ V}$$

$$= 20 \left(1 - e^{\frac{-t}{4}} \right) \text{ V}$$

Therefore, voltage $v_o(t)$ is $\boxed{20 \left(1 - e^{\frac{-t}{4}} \right) \text{ V}}$.

Step 7 of 11

(b)

Consider that the switch is closed for a long time. Redraw the circuit for $t < 0$ as shown in Figure 3.

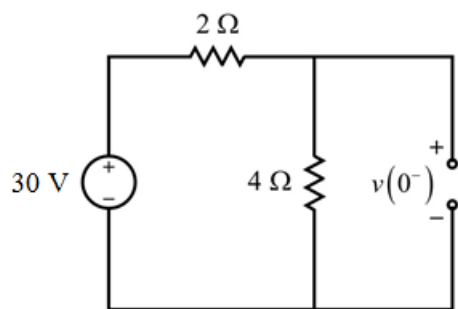


Figure 3

Step 8 of 11

Calculate the initial voltage $v_o(0)$ across the capacitor.

Apply voltage division principle.

$$v_o(0^-) = \frac{4}{2+4}(30)$$

$$= \frac{4}{6}(30)$$

$$= 20 \text{ V}$$

$$v_o(0^-) = v_o(0^+) = v_o(0) = 20 \text{ V}$$

Therefore, the initial voltage $v_o(0)$ is 20 V.

Step 9 of 11

Consider that the switch is opened at $t = 0$. Redraw the circuit for $t > 0$ as shown in Figure 4.

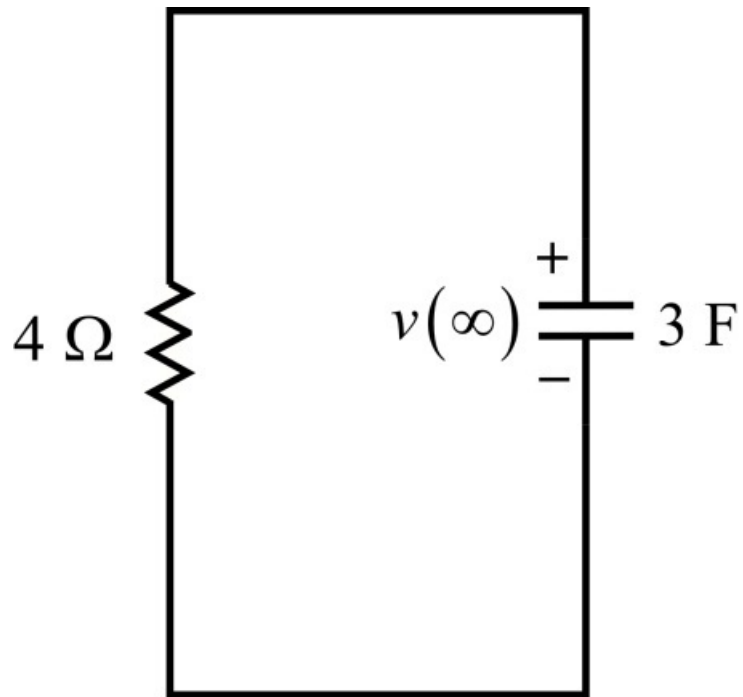


Figure 4

Step 10 of 11

The steady state voltage across the capacitor is,

$$v(\infty) = 0\ \text{V}$$

Calculate the time constant τ .

$$\tau = R_{\text{th}} C$$

Substitute 4Ω for R_{th} , and 3F for C .

$$\begin{aligned}\tau &= 4 \times 3 \\ &= 12\ \text{s}\end{aligned}$$

Step 11 of 11

The expression for the voltage across the capacitor is,

$$v_o(t) = v(\infty) + [v(0) - v(\infty)]e^{\frac{-t}{\tau}}$$

Substitute 20V for $v(0)$, 0V for $v(\infty)$, and 12s for τ .

$$\begin{aligned}v_o(t) &= 0 + [20 - 0]e^{\frac{-t}{12}} \\ &= 20e^{\frac{-t}{12}}\ \text{V}\end{aligned}$$

Therefore, voltage $v_o(t)$ is $\boxed{20e^{\frac{-t}{12}}\ \text{V}}$.